TT-P-0035 (Army-CE)
March 18, 1965

INTERIM FEDERAL SPECIFICATION

PAINT, CEMENTITIOUS, POWDER, WHITE AND COLORS

(FOR INTERIOR AND EXTERIOR USE)

This Interim Federal Specification was developed by the Department of the Army, Corps of Engineers, Washington, D. C. 20315, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE.

1.1 This specification covers a cementitious paint designed for application to interior and exterior surfaces, such as masonry, concrete, stucco, common brick, masonry block and rough plaster, as a decorative, protective and water-repellent coating.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS.

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

SS-C-192 - Cement, Portland.
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; General Specification for Packaging, Packing and Marking.
TT-P-442 - Pigment, Titanium Dioxide (for Protective Coatings).

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 141 - Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing.
Fed. Std. No. 595 - Colors, Standard

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of standards required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

ASTM Standards:

D-659 - Method of Evaluating Degree of Resistance to Chalking of Exterior Paints.

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D-1482 - Method of Test for Color Difference Using the General Electric Spectrophotometer.
D-1495 - Method of Test for Color Difference Using the Color Eye.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.).

3. REQUIREMENTS

3.1 Material. The material shall be in powder form, free from lumps or aggregates that are not easily dispersed with water. The material shall be a good commercial quality suitable for the purpose intended and shall be free of components which may be toxic to personnel under normal conditions of use.

3.2 Composition. The cementitious paint powder shall be as specified in Table I. For dark colors the percentage of titanium dioxide may be varied to a minimum of 2 percent. Color pigments shall be lime-proof and light-fast. Additional components such as dispersion agents, hygroscopic materials, siliceous extenders, thickening agents, water repellents and others may be added to meet the requirements of this specification.

3.3 Quantitative requirements.

3.3.1 The quantitative requirements of the cementitious paint powder shall be as specified in table I.

3.4 Qualitative requirements.

3.4.1 Condition in container. The cementitious powder as received shall not cake, conglomerate or harden as tested in 4.4.1.

3.4.2 Working properties. When blended with water (manufacturer's direction) the material shall have a smooth finish when brushed on concrete panel (Method 2051) at spreading rate of 40 to 60 sq. ft. per gallon. The paint made from the cementitious powder-water mix shall be suitable for application to a damp masonry surface with a fiber brush, a roller or spray equipment. No wetting with water of the applied paint after application shall be necessary when temperature is between 40° and 85° F. and the relative humidity is above 40 percent (see 4.4.2).

3.4.3 Mixing property. The cementitious powder as tested in 4.4.3 shall show no unwetted components after 5 minutes stirring.

3.4.4 Color. The color shall match the standard color agreed upon by buyer and supplier and as indicated in Fed. Std. No. 595 (see Method 4250). If doubt exists as to the match, the color shall show no more than one unit color difference for white paints[1] and no more than two units color difference for dark or mass tones[2] (Method ASTM D-1482, D-1495 or equivalent procedure). (See 6.2).

3.4.5 Color stability (yellowness). The dried film of the material when tested in accordance with Method 6131 shall show no film defects other than very slight yellowing or not more than 0.1 (yellowness of delta N).

3.4.6 Adhesion. The dried film of the material when tested in accordance with Method 6303.1 shall be removed by a (load) weight of not less than 250 grams.

3.4.7 Chalk resistance. There shall be no more than slight chalking of the dried film (degree of chalking No. 6, maximum Figure 1 of ASTM D-659) when exposed in accordance with 4.4.4 and tested in Method 6411 which is also ASTM Method D-659.

3.4.8 Impact resistance. When tested in accordance with paragraph 4.4.5 there shall be no chipping of the paint at a minimum 12-inch pounds load (gardner Mandrel Impact Tester).

3.4.9 Resistance to extreme temperature cycle (120° F. to -10° F.). There shall be no flaking or peeling of the paint film from the masonry substrate, such as concrete panel (2051) or cinder block, after exposure to 5 cycles as tested in accordance with paragraph 4.4.6.

3.4.10 Resistance to Wind-Driven Rain. When tested in accordance with paragraph 4.4.7, the dried film shall pass the test (the required rating being "Excellent").

3.4.11 Weathering (Accelerated). When tested in accordance with paragraph 4.4.8, there shall be no checking, cracking, loss of adhesion, and other defects, and no more than 1-1/2 units color difference for light tints and 3 units for dark or mass colors in the direction of minus delta L, delta a, delta b values from the standard when tested in accordance with ASTM Test Method D-1482, D-1496 or equivalent method.

1 Paints having a reflectance of 80 or more.
2 Paints having a reflectance of less than 80.

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TABLE I

Characteristics	Requirement	
	Minimum	Maximum
Cement, Portland (SS-C-192), type I (White) (% by wt. of powder)	73	--
Titanium Dioxide (TT-P-442, Type III)(% by wt. of powder)	3	5
Hardness (7th to 21st day curing)	30	60
Reflectance, Apparent Daylight (white only)	80	--

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Production inspection of the cementitious powder.
(b) Inspection of preparation for delivery.

4.3 Production inspection.

4.3.1 Sampling and Inspection. Sampling and inspection shall be in accordance with Fed. Test Method Std. No. 141, Section 1031.2.

4.4 Test procedures. The paint (cementitious powder-water mixed) (see 3.4.2) shall be tested in accordance with the following applicable methods of Fed. Test Method Std. No. 141 as indicated in Table II and as herein specified. Failure to pass any test and non-compliance to the requirements shall be cause for rejection of the lot.

4.4.1 Condition in container. Transfer a portion of the powder as received to a 600-ml beaker and examine as specified in 3.4.1.

4.4.2 Working properties. Mix the powder with water (according to manufacturer's direction), brush on a concrete panel (Method 2051) using fiber brush and allow to dry at room temperature for 18 hours. Examine the dried film for compliance as specified in 3.4.2.

4.4.3 Mixing property. Mix 100 grams of the powder with 75 ml of water and stir for 5 minutes. After 5 minutes examine the mixture for compliance as specified in 3.4.3.

4.4.4 Chalk resistance. See weathering (accelerated) test.

4.4.5 Impact resistance. The cementitious paint shall be applied on a concrete panel (Method 2051) and allowed to cure for 21 days at room temperature. Then run the test using Gardner Mandrel Impact Tester and its method. Evaluate for compliance as specified in 3.4.8.

4.4.6 Resistance to extreme temperature cycle (120° F. to -10° F.). Apply the paint (cementitious powder-water mixed) in accordance to manufacturer's direction on a concrete panel (Method 2051) and allow to cure for 7 days at standard conditions. Then expose the coated panel for 5 cycles. Each cycle shall consist of one hour at 120° F., and immediately to one hour at -10° F. At the end of the fifth cycle examine for compliance as specified in 3.4.9.

4.4.7 Resistance to wind-driven rain.

4.4.7.1 Unless otherwise specified the test panels shall be as follows: Four 3 by 7-1/2 by 15-1/2 inch cinder blocks cemented together with mortar compound of 1 part of Portland Cement and 2 parts sand to make a panel approximately 3 by 16 by 32 inches and completely coated with the required system, to conform with the manufacturer's printed directions for surface preparation, mixing, application, coverage and curing.

4.4.7.2 Apparatus. The fundamental parts of the apparatus shall consist of a plexiglas testing box, a source of air under pressure, a manometer measuring in inches, a rubber gasket, clamps and angle irong, and a drain channel.

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4.4.7.2.1 Testing box. The testing box is assembled from a number of transparent plexiglas panels 1/2-inch thick and of the dimensions shown in Figure 1. The fully assembled unit is shown in the photographs, Figures 2 and 2A. The panels are assembled so that all joints are water- and air-tight, a suitable adhesive being employed. For added strength machine screws 5/8-inch long by 3/16-inch - 24 T/I are used along the abutting panel edges. These should be in sufficient number to provide the mechanical strength required, depending upon the maximum pressures to be used. An opening 12 by 24 inches is left on one side of the testing box against which the coated masonry panel to be tested is placed. The top panel has a pair of openings drilled and tapped to receive standard 3/8-inch fittings to which air inlet and manometer lines are connected. These openings should not be less than 3 inches apart to make certain that air inlet turbulences will not affect the manometer readings. A drain opening shall be provided in the lower end of one end panel. This is drilled and tapped to receive a standard 1/2-inch fitting to which a drain hose is connected. A 1/2-inch plexiglas tubing with 1/32-inch holes bored on 1/2-inch centers shall be secured within the testing box, as shown in Figure 1, with the 1/32-inch holes facing the 12 by 24-inch opening in the testing box. One end of the tube shall be sealed, being cemented with epoxy glue against the inside face of one end panel. The opposite end of the tube shall be cemented in place and shall be in line with a water inlet opening drilled and tapped to receive a standard 3/8-inch fitting to which a water supply hose is attached. When the coated test panel is secured in place over the opening in the box, the openings in the plexiglas tubing will permit the full width of the panel to be sprayed with water. An interior plastic brace 1 by 1 by 6 inches will strengthen the front and rear panels. A pair of plastic blocks 2 by 2 by 6 inches are secured to the lower end of the open side of the testing box to prevent tipping of the box when the test panel is attached to the box.

4.4.7.2.2 The source of air delivered into the testing box shall be carefully regulated so that the pressure produced by the water in the box interior can be automatically supplemented as needed. A small air compressor that will automatically supply air may be used. *

4.4.7.2.3 The manometer may be like Model #94125 of Central Scientific Co., Chicago, Illinois, but measuring in inches. The tube shall contain water as this test uses pressure registered in inches of water.

4.4.7.2.4 The rubber gasket shall be similar to that used on refrigerators, approximately 3/4-inch wide and cemented to the edges of the testing box opening by epoxy glue to provide a water- and air-tight seal.

4.4.7.2.5 The clamps and angle irons are shown in the photograph, Figure 2, and shall be of a size and strength to securely clamp the test panel to the testing box and in air- and water-tight engagement with the rubber gasket.

4.4.7.2.6 A drain channel is necessary so that any water passing through the coated masonry can be measured.

4.4.7.2.7 When a masonry panel is to be tested, the rubber gasket shall be first coated with vaseline. The panel shall then be placed against the gasket and over the opening in the testing box. The angle irons shall be positioned, as shown in Figure 2, and clamped tightly so that the seal between the panel and box is air- and water-tight. The unit is then ready for conducting the test.

4.4.7.3 Procedure.

4.4.7.3.1 The water shall be started through the spray pipe so that it strikes the coated panel and flows evenly down all the coated surface at a uniform rate between 40 and 60 gallons per hour, as measured from the water discharged. The exact amount of water discharged per minute shall be reported. The air supply shall be adjusted to supplement the pressure supplied by the water so that 5 inches of water pressure is produced inside the apparatus and as registered by the manometer.

4.4.7.3.2 Leaks shall be checked around the junction of the test panel and the apparatus. Any leaks shall be corrected.

4.4.7.3.3 The flow of water shall be adjusted so the water outlet of the apparatus is always covered. (Maintaining about 4 inches of water in the bottom of the apparatus will do this.)

4.4.7.3.4 The test shall be allowed to run for 8 hours, unless water comes through at such a rate that the coating is declared unsatisfactory before the 8 hours specified.

4.4.7.3.5 The attached graph, Figure 3, shall be consulted for the interior box pressure required for the desired mile-per-hour wind pressure. Unless otherwise specified, the test is made at 5 inches water pressure which is equivalent to 98 mile-per-hour wind pressure.

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* One suitable air pressure regulator is made by Central Scientific Company, Chicago, Illinois, Model #13913.

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4.4.7.4 Observations. Continual observations of the specimens shall be made for two hours after starting the test, after which the observer shall inspect the wall at increasingly longer intervals. The following observations shall be made during the test:

1. Time required for the appearance of moisture (dampness) on the backs of the wall (the uncoated part).

2. Time required for the appearance of visible water on the backs of the wall.

3. Time required for a water leakage to be noted.

4. Maximum rate of leakage, if any.

5. Extent of damp areas on the backs of the walls, including that produced by the capillary rise of moisture from water.

4.4.7.5. Wall performance ratings.

1. Excellent (E). No water visible on the back of the wall at the end of 8 hours. Not more than 25 percent of the wall area damp at the end of 8 hours. No leaks through the wall at the end of 8 hours.

2. Good (G). No water visible on the back at the end of 8 hours. Not more than 50 percent of the wall area damp at the end of 8 hours. No leaks through the wall at the end of 8 hours.

3. Fair (F). Water visible on back of wall in more than 2 and less than 8 hours. Rate of leakage through the wall less than 1 liter per hour at the end of 8 hours.

4. Poor (P). Water visible on the back in 2 hours or less. Rate of leakage less than 5 liters per hour at the end of 8 hours.

5. Very Poor (VP). Rate of leakage through the wall equal to or greater than 5 liters per hour at the end of 8 hours.

4.4.8 Accelerated Weathering. Panels shall be prepared in accordance with Method 2051, Procedure C, and coated to a dry film thickness of 25 mils plus or minus 2.5 mils. The applied coating shall be allowed to dry for 7 days at standard conditions. Accelerated weathering shall be tested for 330 hours in an Atlas Type XW Sunshine Arc Weatherometer which is in accordance with Method 6151, or for 500 hours in an Atlas Type DMC enclosed Violet Carbon Arc Weatherometer or equivalent, operated in accordance with Method 6152 using cycle of 102 minutes of light and 18 minutes of light and demineralized water.

4.4.9 Hardness. Brush the paint on concrete panel (Method 2051) at spreading rate of 40 to 60 sq. ft. per gallon and allow to dry at standard conditions. At the end of seven days run the test using Barber-Coleman Impressor, making five readings. Run another set of tests after fourteen days and twenty-one days respectively. At the end of the tests evaluate for compliance as specified in Table I.

TABLE II

Characteristics	Requirement Reference	Applicable Tests: Fed. Test Method Std. No. 141	Applicable Tests: Paragraph Reference
Condition in container	3.4.1		4.4.1
Working properties	3.4.2		4.4.2
Mixing property	3.4.3		4.4.3
Color	3.4.4	4250 [1]	
Color Stability (Yellowness) White and near-white only	3.4.5	6131 [1]	
Adhesion	3.4.6	6303.1 [1]	
Chalk Resistance	3.4.7		4.4.4
Impact Resistance	3.4.8		4.4.5
Resistance to extreme temperature cycle	3.4.9		4.4.6
Resistance to Wind-driven Rain	3.4.10		4.4.7
Weathering (Accelerated)	3.4.11	6151 or 6152	4.4.8
Cement (Portland)	Table I	7241	
Titanium Dioxide	Table I	7081	
Hardness	Table I		4.4.9
Reflectance (Daylight)	Table I	6121 [2]	

Footnote: [1] Brush the paint on concrete panel (Method 2051) at spreading rate of 40 to 60 sq. ft. per gallon.

[2] Apply the paint as in Method 6121 on Morest chart or equivalent.

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5. PREPARATION FOR DELIVERY.

For civil agency requirements the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 Packaging. Packaging shall be level A, B or C as specified.

5.1.1 Levels A and B. The cementitious powder specified shall be furnished in 5 pound, 25 pound or 50 pound quantities or in quantities used in commercial practice or as specified.

5.1.1.1 Twenty-five and fifty pound quantities. The twenty-five or fifty pound quantities shall be packaged in sacks conforming to UU-S-48, grade L/W or 8 L/W or 14 L/W. Dipped ends are not required. Contractors may use their own commercial moistureproof sacks if they certify that their sacks are equal to or surpass the above-specified multiwall sacks. When specified, the material shall be packed in fiber drums conforming to type I, grade a, PPP-D-723. Drums shall be furnished with a moisture vapor barrier, lining, interior coating or polyethylene bag giving the required protection for hygroscopic materials. Five pound quantities shall be packaged in 1-gallon paint cans.

5.1.2 Level C. The product specified shall be packaged in accordance with good commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified.

5.2.1 Levels A and B.

5.2.1.1 Five pound, twenty-five pound and fifty pound quantities. The product in five, twenty-five and fifty pound quantities packaged as specified in 5.1.1.1 shall require no further packing.

5.2.2 Level C.

5.2.2.1 The product shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. The cementitious paint covered by this specification is intended for interior and exterior masonry surfaces as a decorative, protective and water-repellent coating.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color required (see 3.4.4).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Quantity required (see 5.1).

6.3 Reference standards. Unless other provisions are made by the bureau, department, or other Government activity concerned, the contracting officer should arrange to furnish the supplier with the necessary reference color (Fed. Std. No. 595).

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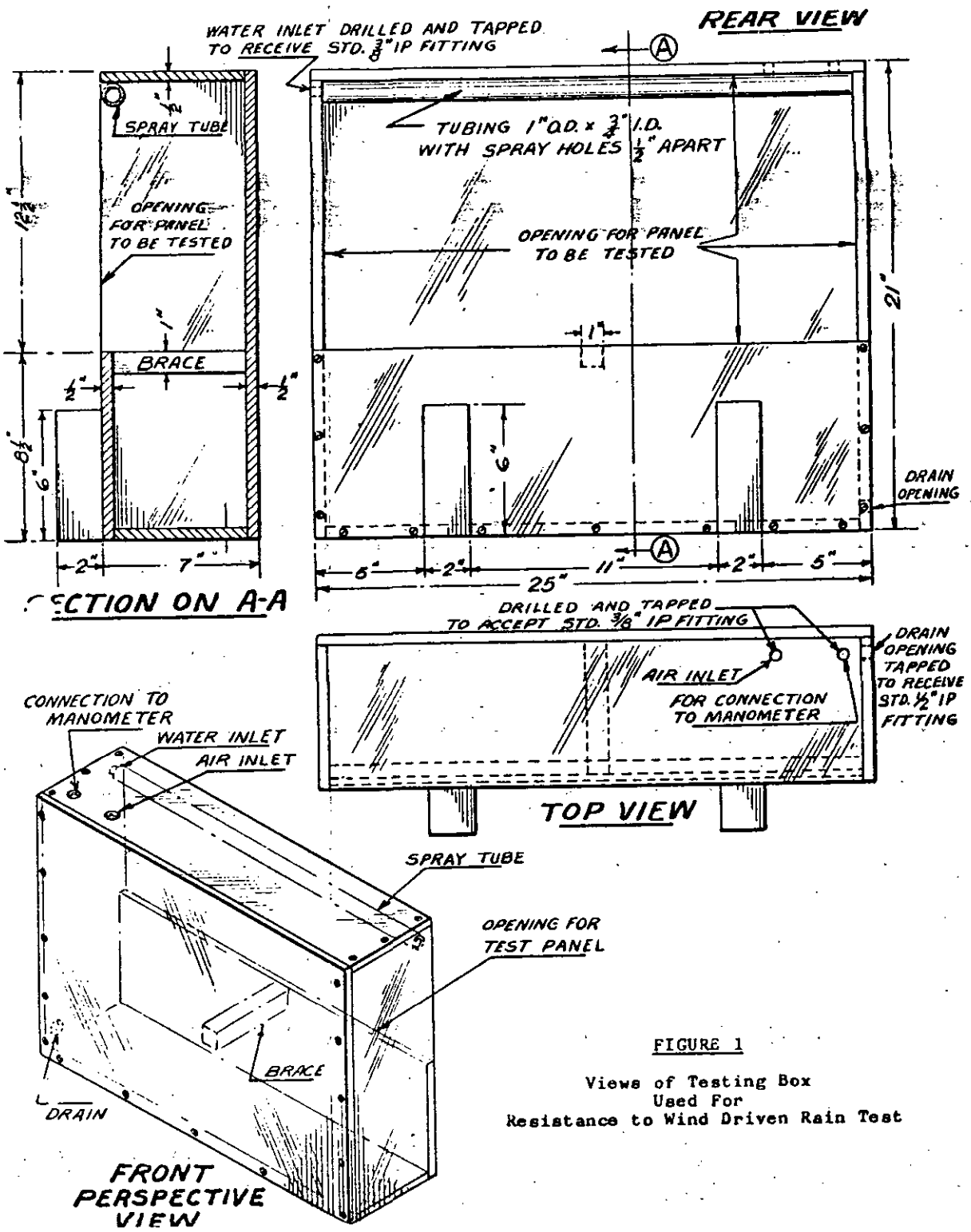


FIGURE 1

Views of Testing Box
Used For
Resistance to Wind Driven Rain Test

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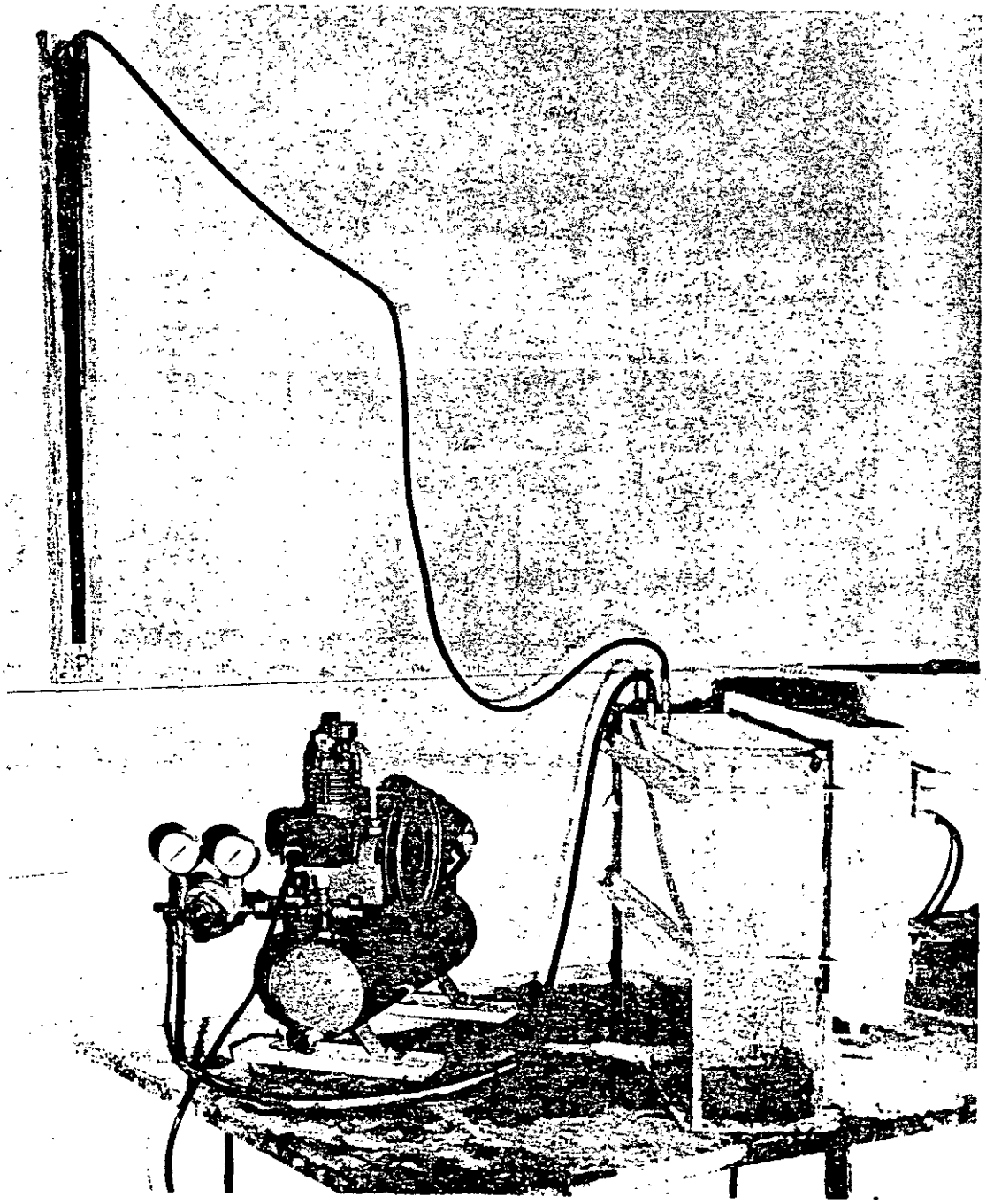


FIGURE 2 *

Photograph of Equipment Used for Resistance to Wind-Driven Rain Test

* May be secured from Moore Research Laboratories, Inc., 2221 Kansas Ave., Silver Spring, Maryland (Tel. No. 301-585-6171.)

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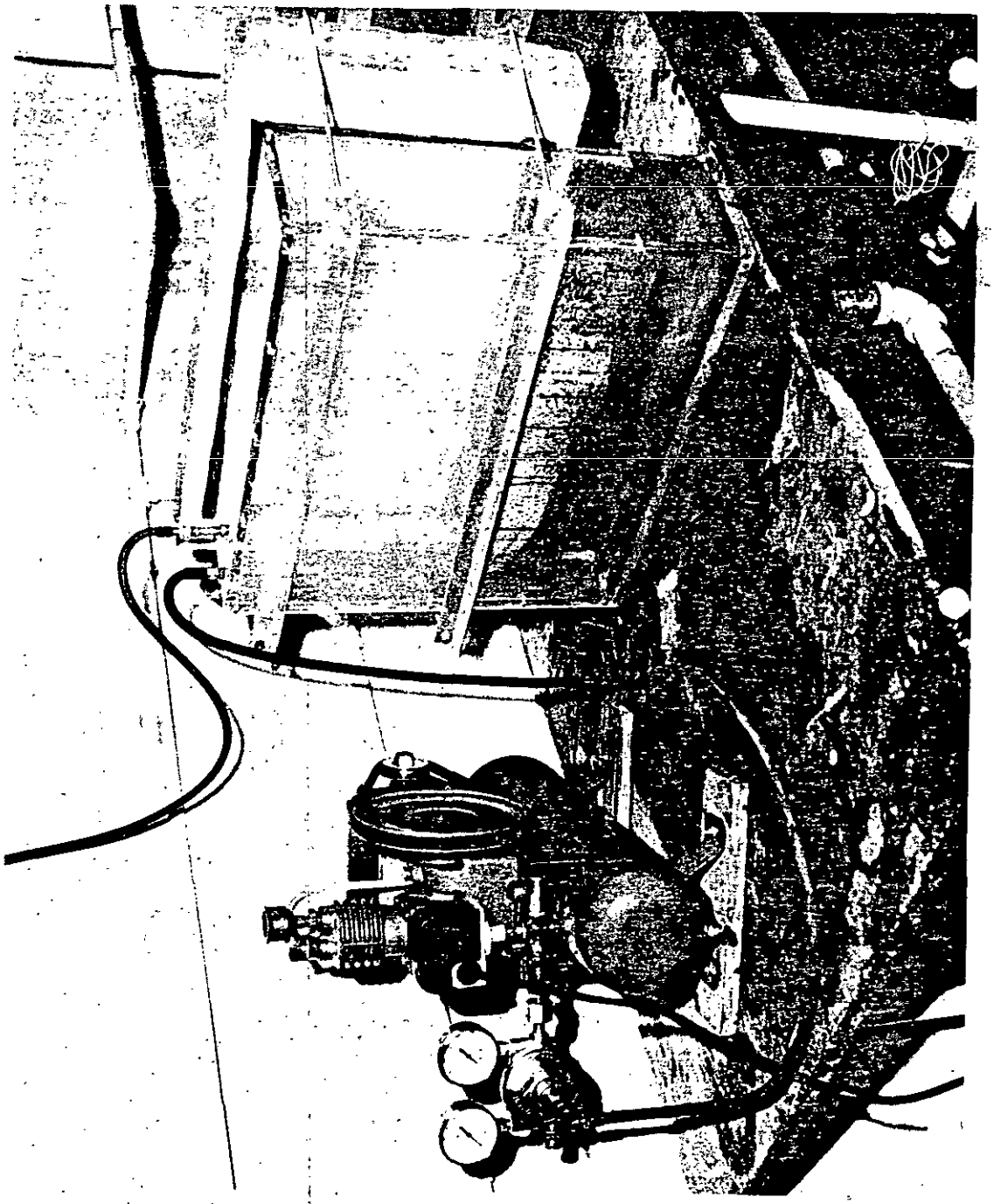
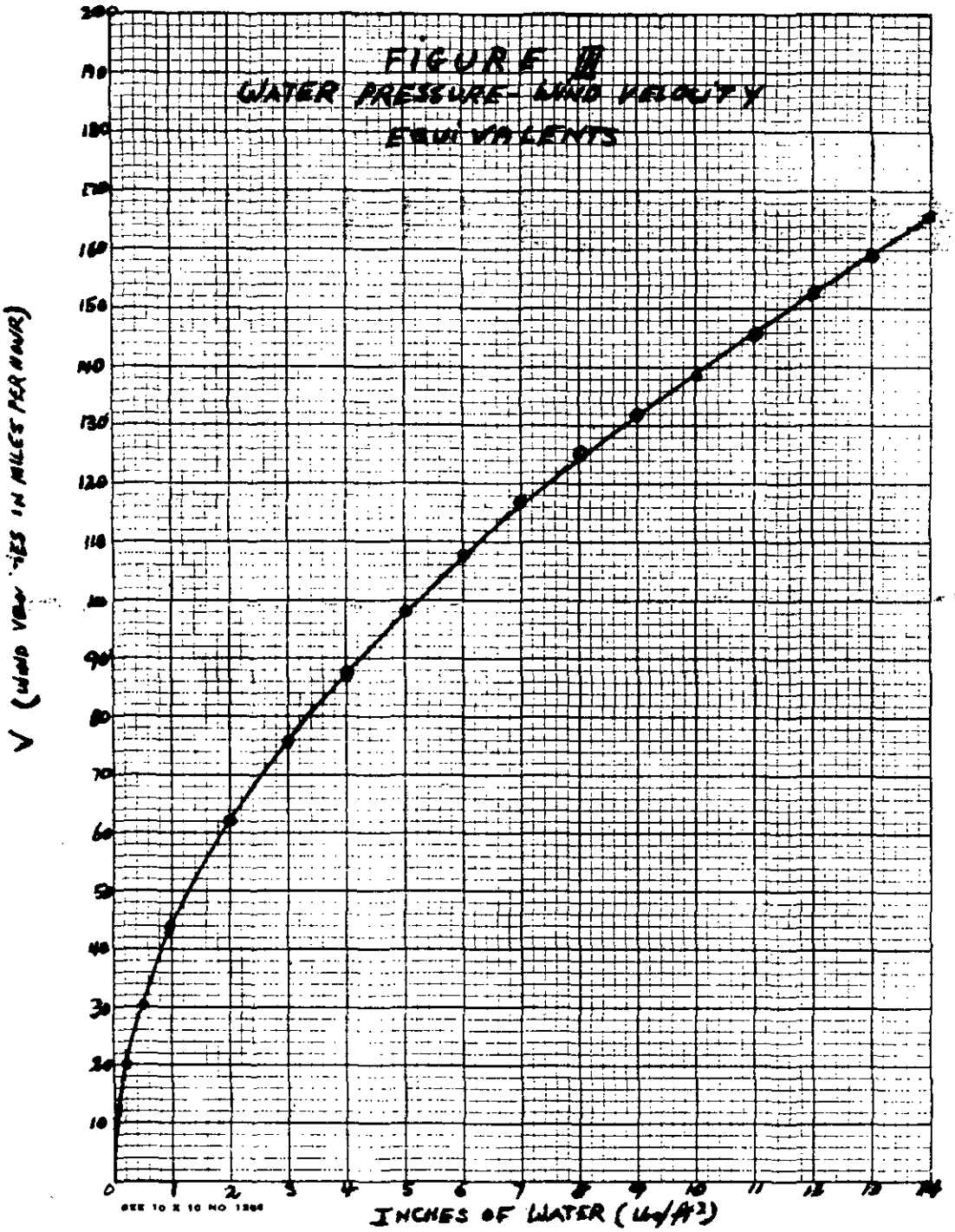


FIGURE 2A *

Photograph of Equipment Used for Resistance to Wind-Driven Rain Test
(Side View)

* May be secured from Moore Research Laboratories, Inc., 2221 Kansas Ave., Silver Spring, Maryland (Tel. No. 301-585-6171.)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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